



Securing understanding of place value in metric units

Task A: Convert metric measures using place value

1) Convert 7 km into:

a) metres
7000 m

b) centimetres
700 000 cm

c) millimetres
7 000 000 mm

	kilo			metres		centi	milli	
	7	0	0	0	0	0	0	

2) Convert 425 ml into litres.

		kilo			litres		centi	milli		
					0	4	2	5		

0.425 litres

3) What is the mass of this pear in kilograms?



163 g

		kilo			grams		centi	milli		
		0	1	6	3					

0.163 kg

4) What is the capacity of this teapot in ml?



1.15 litres

		kilo			litres		centi	milli		
					1	1	5	0		

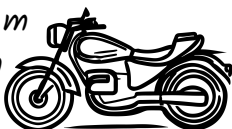
5) Match the objects with all the possible measures.



mass of a drum
6.3 kg



capacity of a jug
630 ml



distance a motorcycle travels
6 300 000 mm

a) 0.0063 g

b) 630 km

c) 6.3 litres

d) ~~6300 g~~

e) ~~6.3 km~~

f) ~~0.63 litres~~

g) 63 000 litres

h) 630 g

i) ~~6300 m~~



Task B: Use conversion to compare different measures

1) Which of these cats is the longest?



107 cm
1070 mm
This cat is the longest.



1009 mm
1009 mm



1.06 m
1060 mm

2) Which of these dogs has a greater mass?



6500 g
6500 g



7 kg
7000 g
This dog has the greater mass

3) The 1.3 litre bottle has a larger capacity. What number could go onto the smaller bottle?

Any number less than 1300 ml



4) A different car has a greater mass than this car.



2300 kg

Write down a possible mass for this other car in grams.

Any number greater than 2 300 000 g

5) Place these distances in order of size, smallest first.

a

1.8 km
1800 m

b

7500 cm
75 m

c

180 m
180 m

d

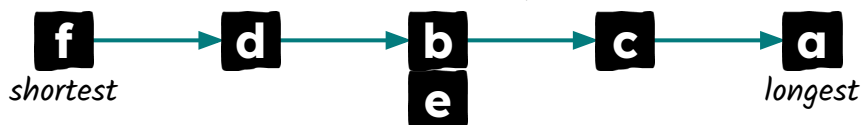
75 mm
0.075 m

e

75 m
75 m

f

1.8 cm
0.018 m



6) Fill in each blank with any number that makes the descriptions underneath correct.

900 g

lighter

kg

heavier

Any number greater than 0.9 kg

kg

lighter

18 500 g

heavier

Any number less than 18.5 kg

7.5 litres

smaller

ml

larger

Any number greater than 7500 ml



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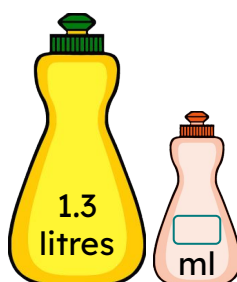


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75 mm

0.075 m

e

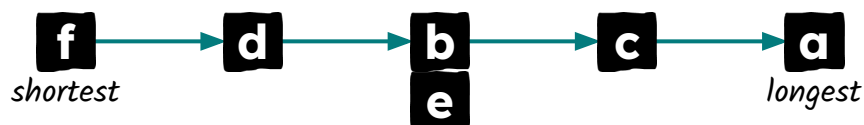
75 m

75 m

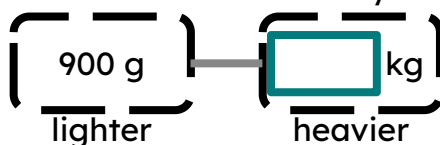
f

1.8 cm

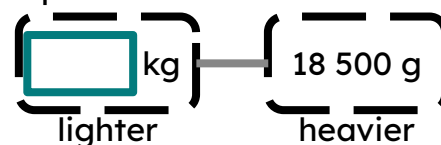
0.018 m



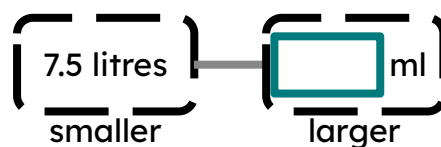
6) Fill in each blank with any number that makes the descriptions underneath correct.



Any number greater than 0.9 kg



Any number less than 18.5 kg



Any number greater than 7500 ml



Task C: Check if conversions of real examples are sensible

1) The mass of the bike below is 14 570 g.
Which of the values below also represents this bike?



1.457 kg

14.57 kg

145.7 kg



2) The capacity of this bathtub when full is 182 000 ml.

Which of the values below also represents the bathtub?

18 200 litres

182 cl

18 200 cl

18.2 litres

182 litres

3) This ship is 0.29 km long and has a mass of 113 000 000 kg.
Fill in the blanks below with the correct numbers and metric units.



290

m

29 000

cm

113 000 000 000

g

(that is 9 zeros)

4) Write down all the possible units of measure that could go in the blanks to make the descriptions underneath correct.



length of Tower Bridge, London

24 400 cm

Shorter
244 m

225

km

Longer



the height of Izzy

16.5

cm
mm

Shorter

0.0017 km

Taller

1.7 m = 170 cm

